

Behavioral Innovation in the Middle and Upper Paleolithic



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The later Middle and Upper Paleolithic saw the widespread introduction of new technologies, both those related to food acquisition and processing, and those related to other aspects of life. As shown earlier in this module, Sally McBrearty and Alison Brooks, in a highly influential [2000 article \(https://www.sciencedirect.com/science/article/pii/S0047248400904354?via%3Dihub\)](https://www.sciencedirect.com/science/article/pii/S0047248400904354?via%3Dihub), identify the sequential introduction of these innovations beginning around 300,000 years ago in Africa. These technologies then spread along with *Homo sapiens* into Eurasia. Some of these technologies are focused on symbolic behavior, including the use of pigments; specialized, rough grinding stones could be used to create paints used in body adornment and the creation of art. At the same time, advanced stone technologies, such as microblade, allowed the creation of specialized tools used later in grain harvesting. Barbed points and bone tools were used extensively in fishing alongside other technologies, such as fiber nets, of which we have only limited evidence due to the selective degradation of organic materials.

At the same time, anatomically (and also, eventually, behaviorally), modern humans spread across Africa, Eurasia, and, by c. 60,000 years ago, Australia. By c. 30,000 years ago, we replaced all other species of archaic humans, including Neanderthals and Denisovans, becoming the only human species on the planet. Why Neanderthals went extinct is still hotly debated, but it is likely that their large body size, while advantageous in the cold climates of the Pleistocene Northern Hemisphere, were ultimately an evolutionary disadvantage against modern humans, who could use clothing to stay as warm in cold climates without the metabolic cost of extra body mass. Another contributing factor may have been the advanced technological tool kit and complex systems of social learning that were certainly present among modern humans during this period—although Neanderthals likely shared at least some of these advanced behaviors, possibly including advanced language, hunting equipment, and symbolic behavior. It is likely, however, that modern humans were able to make better or more efficient use of limited food resources during the Late Pleistocene, which eventually enabled them to outcompete Neanderthals and replace them.